

Reference??

Integer
Modulo
Modulo
modulo
mod
 $\mathbb{Z}$
 $\mathbb{Z}$

(i.e.,
modulo q)
 $\{0, 1, 2, 3, \dots, q-1\}$

Modulus:

Modulo
Con-
gru-
ence
($\equiv$):

$$\begin{pmatrix} a \\ b \\ q \\ \bar{a} \\ b \end{pmatrix} \equiv \begin{pmatrix} a \\ b \\ q \\ \bar{a} \\ b \end{pmatrix} \pmod{q}$$
$$q \equiv 5 \pmod{12}$$
$$\begin{aligned} & 12 \bmod 7 \\ & 5 \bmod 7 \\ & = \end{aligned}$$
$$5 \equiv 12 \pmod{7}$$
$$a \equiv b \pmod q$$
$$\frac{q}{a} \equiv \frac{q}{b} \pmod{q}$$
$$\begin{pmatrix} a \\ q \\ b \\ a \end{pmatrix}$$
$$\frac{q}{a} = \frac{q}{b} \bmod$$

Congruence

vs.
Equal-
ity:

$$a \equiv b \pmod{q}$$
$$\frac{a}{b} +$$
$$\frac{0+}{k \cdot q}$$
 q_{ka}
$$\begin{matrix} a \\ b \\ q \end{matrix}$$
$$\begin{matrix} q \\ a \\ b \end{matrix}$$

59 =

$$\begin{matrix} 5 \\ 12 \\ 7 \\ 5 \end{matrix} \begin{matrix} \equiv \\ \text{mod} \\ \iff \\ \equiv \end{matrix}$$
$$\frac{5}{7} \frac{12}{(-1)}.$$

7 ?? .1

Properties

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of
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tions

x Addit

$$\begin{array}{c} \overline{a} \\ b \\ q \end{array} \equiv \text{mod} \begin{array}{c} \overline{a} \\ b \\ q \end{array} \begin{array}{c} \overline{a} \\ b \\ q \end{array} \begin{array}{c} \overline{a} \\ b \\ q \end{array}$$
$$\begin{array}{c} a+ \\ x \equiv \\ b+ \\ x \bmod \end{array}$$